

Equivalence of the path-integral theory of spinning particles and the topological nonlinear σ model in $d=2$ dimensions

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We show that the path-integral theory of a spinning particle with spin S is equivalent to the topological $O(3)$ nonlinear σ model (TNLSM) in $1+1$ Euclidean dimensions. We prove this equivalence in two different ways. In particular, we use stochastic quantization of the theory of spin to show the identity of the generating functionals of correlation functions of both theories. Also, using canonical quantization of the TNLSM, we show that its ground-state wave function coincides with the Feynman weight for the path-integral theory of the spin. In addition, we give a full classification of the invariants of the TNLSM. The invariants are computed explicitly in the case of the sphere.

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I. INTRODUCTION

The study of topological quantum field theories (TQFT's) has clarified the role and properties of topological invariants for low-dimensional manifolds and their connection with quantum field theory. Most of the work on this subject has concentrated on the study of the topological invariants in topological Yang-Mills (TYM) theory in four dimensions and in Chern-Simons (CS) theory in three dimensions [1]. In particular, this approach has yielded an explicit construction of Donaldson's invariants, Jones' polynomials, etc. Apart from this mathematical interest, TQFT's are also appealing from the point of view of fundamental physics. It has recently been suggested that they may describe a phase of unbroken diffeomorphism invariance in quantum gravity [2]. Also, the study of Chern-Simons theory has given a relation with rational conformal field theories in two Euclidean dimensions.

In this paper we will consider the role of topological invariance in quantum mechanics. We will study a problem in $0+1$ dimensions which is the analogue of Chern-Simons theory in $2+1$ dimensions: the path-integral theory of spin [3–7]. This is perhaps the simplest topological theory. In fact, the path-integral theory of a spinning particle is just the theory of the Berry phase [8]. More specifically, we show that the coherent-state path integral for a particle with spin S is equivalent, *via* stochastic quantization, to the topological $O(3)$ nonlinear σ model (TNLSM). That is, a nonlinear σ model defined on a two-dimensional manifold M with an action proportional to the topological charge of the smooth maps of $M \rightarrow S^2$. To be more precise, what we prove is the identity between the generating functionals of the correlation functions of the stochastically quantized theory of spin and the TNLSM. Our result is the analogue in quantum

mechanics of the connection that exists between Chern-Simons theory in $d=3$ and topological Yang-Mills (TYM) theory in $d=4$ dimensions [9–12]. Here too, this connection is established by using stochastic quantization.

However, a completely different approach has also revealed the connection between TYM and CS theories. Using canonical quantization, Horowitz has proven [13] that a TQFT in $S^3 \times R$, which is intimately connected to TYM, has a unique quantum state which can be expressed as a wave function on the space of the gauge fields in the form $\Psi = \exp(ig\mathcal{S}_{CS})$, where g is a parameter and $\mathcal{S}_{CS}$ is the CS action. Following the same procedure (i.e., canonical quantization treating constraints in the manner of Dirac) we also construct in the present paper the wave function of the TNLSM and show that it can be written as $\exp(i\frac{1}{2}\mathcal{S})$ where $\mathcal{S}$ is the action for the spinning particle and k an integer. In other words, the wave function of the TNLSM is the same as the Feynman weight of the path-integral theory of the spin. In a sense, we show that the TNLSM is a universal theory of spin which includes all possible representations of $SU(2)$.

The paper is organized as follows. In Sec. II we present the stochastic quantization of the theory of a particle with spin S . Using the Parisi-Wu formalism we show that the generating function for the correlation functions of this theory is, in stochastic quantization, equivalent to the partition function of the topological nonlinear σ model. The equivalence is proven in Sec. III. We also show that the partition functions as well as all expectation values of all the operators in both theories are topological invariants and independent of the coupling constants. In Sec. III we also give a classification of all the invariants of the TNLSM. In Sec. IV we discuss the canonical quantization of the TNLSM and find that its wave functions coincide with the Feynman weights of

the coherent-state path integral for the spin. We also argue that the TNLSM is a theory of all the representations of $SU(2)$. In Sec. V we present an explicit computation of the topological invariants. We give a complete solution of this problem for the case in which the base manifold $M=S^2$ where we show that the only nonvanishing topological invariant is equal to $\pm 4\pi$. In the Appendix we give a summary of the coherent-state path integral for spin.

II. THE STOCHASTIC QUANTIZATION OF A SPINNING PARTICLE

In this section we will develop the stochastic quantization approach for the path-integral formulation of a particle with spin S . In the Appendix we show that the path integral for spin can be described in terms of a vector field $n^a(t)$ ($a=1,2,3$). The action for this theory (both in real and imaginary time) is given by

$$\mathcal{S}^{d=1} = S \int dt \int d\lambda \mathbf{n} \cdot (\partial_t \mathbf{n} \times \partial_\lambda \mathbf{n}), \quad (2.1)$$

$$\mathbf{n}^2 = n^a(t, \lambda) n^a(t, \lambda) = 1, \quad n^a(t, 1) = n^a(t). \quad (2.2)$$

Here t is time ($t \in S^1$) and λ is a parameter ($\lambda \in [0, 1]$) such that, for $S=1$, $\mathcal{S}^{d=1}$ gives the area of a cap Σ in S^2 with S^1 as its boundary, $S^1 = \partial \Sigma$. The equations of motion arising from Eq. (2.1) are independent of λ and read

$$\frac{\delta \mathcal{S}^{d=1}}{\delta \mathbf{n}} = S \mathbf{n} \times \partial_t \mathbf{n} = 0, \quad (2.3)$$

together with the constraint

$$\mathbf{n}^2 = 1. \quad (2.4)$$

In this section we will develop the stochastic quantization procedure for the theory with action $\mathcal{S}^{d=1}$ given in Eq. (2.1). The conventional procedure is the following: given an action, a Markov process in “Monte Carlo time” τ for the field configurations is constructed *via* a Langevin equation. If the stochastic process is purely relaxational, one is guaranteed the existence of an invariant equilibrium probability distribution for the fields. This distribution is the Gibbs distribution of the system at equilibrium. In quantum mechanics, the Gibbs measure is interpreted as the Feynman weight of the path integral (in imaginary time). We will follow this approach despite the fact that (a) the Feynman weight has a factor of i (in both, real and imaginary time) and (b) the action $\mathcal{S}^{d=1}$ is a functional of the first derivative in time of $\mathbf{n}$. Although the convergence of this process is not guaranteed, we will proceed formally and show *a posteriori* (in Sec. IV) that the invariant distributions are the correct ones. An alternative procedure would consist to modify the action $\mathcal{S}^{d=1}$ by the addition of a regulating term which is second order in derivatives [14]. If the coefficient of this term lies in the appropriate range, the modified Langevin equation will have convergent solutions. The effective theory in $d=2$ will no longer be topological since the regulating term contains dimensional parameters. At the end of the calculation, the regulator should be removed yielding the effective topological action which we derive

below. This problem is the analogue in stochastic quantization of a formal difficulty which arises in the *path-integral* quantization of theories which have first-order derivatives in their action, e.g., coherent-state path integrals. In the coherent-state path-integral quantization it is well known that there are no *continuous* solutions to the classical equations of motion. These path integrals are regulated by means of a similar procedure [15]. Notice, however, that these issues are important only when these theories are quantized semiclassically. From now on, we will ignore these subtleties and proceed under the assumption that a regulating method has been adopted.

Let us proceed to the stochastic quantization of the model defined by action of Eq. (2.1). Within this approach one introduces a stochastic time τ , ($\tau \in [-T, T]$) and postulates a Langevin equation for the field $\mathbf{n}$:

$$\frac{\partial \mathbf{n}}{\partial \tau} = - \frac{\delta \mathcal{S}^{d=1}}{\delta \mathbf{n}} + \boldsymbol{\eta}, \quad (2.5)$$

with $\boldsymbol{\eta}$ a Gaussian noise. The constraint Eq. (2.4) demands that $\boldsymbol{\eta}$ should be perpendicular to $\mathbf{n}$, i.e., $\boldsymbol{\eta} \cdot \mathbf{n} = 0$. More generally, we will consider noise functions $\boldsymbol{\eta}(t, \tau)$ such that the solution $\mathbf{n}(t, \tau)$ of the Langevin equation (2.5) satisfies the constraint $\mathbf{n}^2 = 1$.

Denoting by $\langle \rangle_\eta$ the stochastic expectation values, it is possible to prove [16] the identity

$$\lim_{\tau \rightarrow \infty} \langle n_\eta^{a_1}(t_1, \tau) \cdots n_\eta^{a_g}(t_g, \tau) \rangle_\eta = \langle 0 | T n^{a_1}(t_1) \cdots n^{a_g}(t_g) | 0 \rangle. \quad (2.6)$$

In Eq. (2.6) n_η^a denotes a solution of Langevin equations giving $\mathbf{n}$ as a functional of $\boldsymbol{\eta}$ and the right-hand side (RHS) corresponds to the time-ordered expectation value for the one-dimensional quantum field theory with dynamics defined by action (2.1).

Within the path-integral approach, stochastic expectation values are computed using a Gaussian measure

$$d\mu[\boldsymbol{\eta}] = \exp \left[-\frac{1}{2} \int_M dt d\tau \boldsymbol{\eta}^2 \right] \delta(\mathbf{F}^{-1}[\boldsymbol{\eta}]^2 - 1) D\boldsymbol{\eta}, \quad (2.7)$$

where M is the two-dimensional compact manifold constructed from $S^1 \times [-T, T]$ by choosing appropriate boundary conditions at $\pm T$. We shall discuss different choices of M later on. The delta function in the measure of Eq. (2.7) denotes the restriction to the set of functions $\boldsymbol{\eta}$ for which the Langevin equation has a solution compatible with the constraint. The functional $\mathbf{F}$ will be defined below in Eq. (2.8).

Let us rewrite (2.5) in the form

$$\mathbf{F}[\mathbf{n}] \equiv \partial_0 \mathbf{n} + \mathbf{n} \times \partial_1 \mathbf{n} = \boldsymbol{\eta}, \quad (2.8)$$

where we have denoted

$$\partial_0 \equiv \frac{1}{S} \partial_\tau, \quad \partial_1 \equiv \partial_t, \quad (2.9)$$

and redefined $\boldsymbol{\eta} \rightarrow S\boldsymbol{\eta}$. We are now ready to find the connection between the stochastically quantized version of

the model defined by action (2.1) and a two-dimensional topological model. To this end, we start from the generating function for the $d=1$ model, hereafter called the “stochastic partition function,” written in the form

$$Z_{\text{stoch}}^{d=1} = \int D\eta \exp \left[-\frac{S}{2} \int_M d^2x \eta^2 \right] \quad (2.10)$$

with Eqs. (2.4) and (2.8) as constraints and we have absorbed in the definition of the measure $D\eta$ the restriction of Eq. (2.7). We have also used the notation $d^2x = dx^0 dx^1$.

We shall make the constraints explicit by inserting in (2.10) the “resolution of the identity:”

$$1 = \int D\mathbf{n} \delta(\eta - \mathbf{F}[\mathbf{n}]) \Delta[\mathbf{n}] . \quad (2.11)$$

When inserted in the stochastic partition function Eq. (2.10) the restriction of Eq. (2.7) yields the measure $D\hat{\mathbf{n}}$, which is the invariant measure on the sphere

$$D\hat{\mathbf{n}} = D\mathbf{n} \delta(\mathbf{n}^2 - 1) \quad (2.12)$$

and

$$\Delta[\mathbf{n}] = \det \left[\frac{\delta \mathbf{F}}{\delta \mathbf{n}} \right] . \quad (2.13)$$

In order to compute Δ , let us consider a variation $\delta \mathbf{n}$ such that the transformed vector $\mathbf{n}', \mathbf{n}' = \mathbf{n} + \delta \mathbf{n}$, also satisfies constraint (2.2):

$$\delta \mathbf{n} = \mathbf{n} \times \delta \epsilon , \quad (2.14)$$

with $\delta \epsilon$ an arbitrary infinitesimal variation. Let $\delta \mathbf{F}$ be an infinitesimal variation of $\mathbf{F}$:

$$\delta \mathbf{F} \equiv \mathbf{F}[\mathbf{n}'] - \mathbf{F}[\mathbf{n}] . \quad (2.15)$$

In components we can write

$$\delta F^a = \frac{\delta F^a}{\delta n^b} \epsilon_{bpq} n^p \delta \epsilon^q . \quad (2.16)$$

Thus, Δ is given by

$$\Delta = \det \mathcal{M}^{ab} , \quad (2.17)$$

where the matrix $\mathcal{M}^{ab}$ is

$$\mathcal{M}^{ab} = \frac{\delta F^a}{\delta n^b} \epsilon_{pqb} n^q . \quad (2.18)$$

We now introduce ghosts and antighosts (c and $\bar{c}$, with ghost number $+1$ and -1 , respectively) in order to write Δ in (2.17) as a path integral:

$$\Delta = \int D\bar{c} Dc \exp \left[\int \bar{c}^a \mathcal{M}_{ab} c^b d^2x \right] . \quad (2.19)$$

Inserting (2.11) in (2.10) and using (2.19), we then get

$$Z_{\text{stoch}}^{d=1} = \int D\hat{\mathbf{n}} D\eta \delta(\eta - \mathbf{F}[\mathbf{n}]) D\bar{c} Dc \times \exp \left[-\frac{S}{2} \int_M (\eta^2 + \bar{c} \mathcal{M} c) d^2x \right] . \quad (2.20)$$

We can integrate out the Gaussian noise using the δ func-

tion. It will also be useful to introduce a Lagrange multiplier $\mathbf{b}$ so that one finally has

$$\begin{aligned} & \int D\eta \delta(\eta - \mathbf{F}[\mathbf{n}]) \exp \left[-\frac{S}{2} \int_M \eta^2 d^2x \right] \\ &= \int D\mathbf{b} \exp \left[\int_M \left[\frac{\mathbf{b}^2}{2} - \sqrt{S} \mathbf{b} \cdot \mathbf{F} \right] d^2x \right] . \end{aligned} \quad (2.21)$$

Thus, we find that the generating function $Z_{\text{stoch}}^{d=1}$ is

$$\begin{aligned} Z_{\text{stoch}}^{d=1} &= \int D\hat{\mathbf{n}} D\mathbf{b} D\bar{c} Dc \\ &\times \exp \left[-\int_M \left[\mathbf{b} \cdot \mathbf{F} - \frac{\mathbf{b}^2}{2S} + \bar{c} \mathcal{M} c \right] d^2x \right] . \end{aligned} \quad (2.22)$$

It is easy to verify that, as usually happens after stochastic quantization [9] the effective action appearing in (2.22) has a Becchi-Rouet-Stora-Tyutin (BRST) invariance,

$$(a) \{Q^\perp, \mathbf{n}\} = \mathbf{n} \times \mathbf{c}, \quad (b) \{Q^\perp, \bar{c}\} = \mathbf{b} , \quad (2.23)$$

$$(a) \{Q^\perp, \mathbf{c}\} = -\frac{1}{2} \mathbf{c} \times \mathbf{c} , \quad (b) \{Q^\perp, \mathbf{b}\} = 0 , \quad (2.24)$$

where $\{Q^\perp, \}$ is a symbol representing a BRST variation:

$$\delta_{\text{BRST}} \mathbf{a} \equiv \lambda \{Q^\perp, \mathbf{a}\} , \quad (2.25)$$

with λ a (global) anticommuting parameter. Of course, $\delta_{\text{BRST}}^2 = 0$, as can be easily seen from Eqs. (2.23) and (2.24).

As usual, the form of transformation law (2.23a) was inferred from (2.14), by associating a ghost $\mathbf{c}$ with the parameter $\delta \epsilon$ which describes the $\mathbf{n}$ variation

$$\delta \epsilon \rightarrow \mathbf{c} . \quad (2.26)$$

Once this is done, Eq. (2.24a) is derived by demanding $\delta_{\text{BRST}}^2 \mathbf{n} = 0$. It should be noted, however, that $\mathbf{n}$ variations are defined modulo rotations parallel to $\mathbf{n}$. Namely, if we consider

$$\delta_{\parallel} \epsilon = \mathbf{n} \delta \rho , \quad (2.27)$$

then

$$\delta \mathbf{n} = 0 . \quad (2.28)$$

We are then facing the problem of a so-called *second-generation gauge invariance* [17] and we have to introduce a ghost ϕ associated with $\delta \rho$:

$$\delta \rho \rightarrow \phi . \quad (2.29)$$

As we did before, we infer the contribution of this invariance to the BRST transformation law for $\mathbf{c}$ from Eq. (2.27):

$$\delta_{\parallel \text{BRST}} \mathbf{c} = \lambda \mathbf{n} \phi . \quad (2.30)$$

It is clear from (2.30) that ϕ is Grassmann even and has ghost number 2.

The complete set of BRST transformations results from modifying (2.23) and (2.24) so as to take into account (2.30). One gets

$$(a) \{Q, \mathbf{n}\} = \mathbf{n} \times \mathbf{c}, \quad (b) \{Q, \bar{\mathbf{c}}\} = \mathbf{b}; \quad (2.31)$$

$$(a) \{Q, \mathbf{c}\} = -\frac{1}{2} \mathbf{c} \times \mathbf{c} + \mathbf{n} \phi, \quad (b) \{Q, \mathbf{b}\} = 0; \quad (2.32)$$

$$(a) \{Q, \phi\} = 0, \quad (b) \{Q, \bar{\phi}\} = b_0, \quad (c) \{Q, b_0\} = 0. \quad (2.33)$$

Here $\bar{\phi}$ is the antighost (with ghost number -2) associated with ghost ϕ and b_0 is a Lagrange multiplier introduced in order to fix the second-generation gauge invariance. We choose the following (natural) condition to fix this symmetry:

$$\mathbf{n} \cdot \mathbf{c} = 0. \quad (2.34)$$

With this, $Z_{\text{stoch}}^{d=1}$ can be written in the form

$$Z_{\text{stoch}}^{d=1} = \int D\hat{\mathbf{n}} D\bar{\mathbf{c}} D\mathbf{c} D\bar{\phi} D\phi D\mathbf{b} Db_0 \exp(-\mathcal{S}_q^1), \quad (2.35)$$

with

$$\mathcal{S}_q^1 = \int_M \{Q, V\} d^2x \quad (2.36)$$

and

$$V = \bar{\mathbf{c}} \cdot \left[\mathbf{F} + \frac{\mathbf{b}}{2S} \right] + \bar{\phi} \mathbf{n} \cdot \mathbf{c}. \quad (2.37)$$

As a summary, we list in Table I the field content (which will be generically called $\Phi = \{\Phi_a\}$) including ghost number and Grassmann parity of the resulting quantized model.

Let us note at this point that, as it appears, $\mathcal{S}_q^1$ is nothing but the gauge-fixing action which should arise by starting from the trivial classical action $\mathcal{S}_{\text{cl}} = 0$ and using as gauge conditions the equation $\mathbf{F}[\mathbf{n}] = 0$ and $\mathbf{c} \cdot \mathbf{n} = 0$ in order to fix the large symmetry (2.14) and the second-generation symmetry (2.27). The fact that $\mathcal{S}_q^1$ takes as a whole the form of a BRST variation has very important consequences. In particular one has

$$\langle T_{\mu\nu} \rangle \equiv -\frac{1}{Z} \frac{\delta Z_{\text{stoch}}^{d=1}}{\delta g^{\mu\nu}} = \left\langle \left\{ Q, \frac{\delta V}{\delta g^{\mu\nu}} \right\} \right\rangle = 0, \quad (2.38)$$

with $g_{\mu\nu}$ the metric on M . In obtaining Eq. (2.38) one has to assume [1] (1) matrix independence of the path-integral measure (which will be called $D\Phi$) in Eq. (2.22) and (2) invariance of $D\Phi$ under BRST transformations (2.31)–(2.33). Notice that Eqs. (2.31)–(2.33) show that

the BRST variations are independent of the metric. Assumption (2) ensures that vacuum expectation values of BRST variations vanish. Concerning assumption (1) one can always define a metric-independent measure not in terms of the original fields Φ but of certain densities $\bar{\Phi}$ obtained by multiplying each Φ_a by appropriate powers w_a of $g = \det g_{\mu\nu}$ [18,19]. In the present case, this amounts to multiplying each field by a $g^{1/4}$ factor [18] since all the fields are Lorentz scalars. One then should write, instead of Eqs. (2.35) and (2.36) [20],

$$Z_{\text{stoch}}^{d=1} = \int D\bar{\Phi} \exp \left[- \int d^2x \{Q, V[\bar{\Phi}, g]\} \right] \quad (2.39)$$

with $\bar{\Phi}_a = g^{1/4} \Phi_a$. One now takes $\{\bar{\Phi}_a\}$ and $g_{\mu\nu}$ as independent variables and then Eq. (2.38) is proven. Let us emphasize that, in this theory, the metric is not a dynamical variable.

Another interesting property related to the peculiar form of action (2.36) is the independence of $Z_{\text{stoch}}^{d=1}$ on the coupling constant. Indeed, for the same reasons making $\langle T_{\mu\nu} \rangle$ vanish, one has

$$\frac{1}{Z} \frac{\delta Z_{\text{stoch}}^{d=1}}{\delta S} = \frac{1}{S^2} \left\langle \int d^2x \{Q, \bar{\mathbf{c}} \cdot \mathbf{b}\} \right\rangle = 0. \quad (2.40)$$

The results of Eqs. (2.38) and (2.40) are characteristic of what is called topological quantum field theory (TQFT) [1] (see Ref. [21] for a review).

In the present case, the resulting TQFT is defined in the two-dimensional manifold M constructed by addition of stochastic time τ to the original one-dimensional space. A similar connection between Chern-Simons theory in $d=3$ dimensions and topological Yang-Mills theory in $d=4$ was established in Refs. [9–11] and investigated in Ref. [12]. In the next section we shall discuss this connection by considering different approaches leading to two-dimensional TQFT's.

III. THE TOPOLOGICAL O(3) NONLINEAR σ MODEL IN $d=2$

Topological quantum field theories can be thought as obtained by BRST fixing of the large (topological) symmetry of some “trivial” action $\mathcal{S}_{\text{cl}}$. Either this action is taken as the topological invariant associated with the model [22] or it is chosen just as a Gaussian action constructed from the instanton equations (Bogomoln’nyi equations) related to the model [23]. In fact, we have already noted that $\mathcal{S}_Q^1$ could be considered as the quantum action arising from the BRST fixing of a large (topological) symmetry defined by Eq. (2.14). What we want to show now is that a partition function equivalent to that given by Eq. (2.35) can be obtained after quantization of a topological classical action, namely that given by the topological charge of the nonlinear σ model [24] in a two-dimensional manifold M :

$$\mathcal{S}_{\text{top}}^{d=2} = \alpha \int d^2x \epsilon_{\mu\nu} \mathbf{n} \cdot (\partial_\mu \mathbf{n} \times \partial_\nu \mathbf{n}). \quad (3.1)$$

This action is of course invariant under transformations (2.14) and one has again the second-generation gauge invariance described by Eqs. (2.27) and (2.28). As in other topological models, it is convenient to choose as gauge

TABLE I. The field content of the stochastically quantized model.

Field	Grassmann	Ghost No.
$\mathbf{n}$	even	0
$\mathbf{c}$	odd	1
$\bar{\mathbf{c}}$	odd	-1
ϕ	even	2
$\bar{\phi}$	even	-2
$\mathbf{b}$	even	0
b_0	odd	-1

fixing for the large topological symmetry the instanton equation associated with (3.1), which is, in fact

$$\mathbf{F}[\mathbf{n}] = 0, \quad (3.2)$$

with $\mathbf{F}$ defined as in Eq. (2.8).

Again, we use as the second gauge-fixing condition the natural one:

$$\mathbf{c} \cdot \mathbf{n} = 0. \quad (3.3)$$

With this, one ends with a partition function in the form

$$Z_{\text{top}}^{d=2} = \int D\Phi \exp(-\mathcal{S}_q^2) \quad (3.4)$$

with

$$\mathcal{S}_q^2 = \mathcal{S}_{\text{top}}^{d=2} + \int \{Q, V_\beta\} d^2x. \quad (3.5)$$

Here

$$V_\beta = \bar{\mathbf{c}}(\mathbf{F} + \beta \mathbf{b}) + \bar{\phi} \mathbf{n} \cdot \mathbf{c} \quad (3.6)$$

and β is a real parameter covering different gauge choices. For instance $\beta=0$ corresponds to the condition $\mathbf{F}[\mathbf{n}]=0$, while the choice $\beta=-1/\alpha$ is such that a piece of the second term on the RHS of Eq. (3.5) cancels out $\mathcal{S}_{\text{top}}^{d=2}$. Notice that $Z_{\text{top}}^{d=2}$ is independent of the value of the gauge-fixing parameter β . Thus, the actual value of the coefficient α in $\mathcal{S}_{\text{top}}^{d=2}$ is unimportant since it can be changed by changing β . A similar analysis to the one we describe below was presented by Iwasaki [25] in a work on topological CP^{n-1} models.

By direct comparison of Eqs. (2.35)–(2.37) with Eqs. (3.4)–(3.6) we get the main result of this section: the stochastic partition function of the theory of a spinning particle is *identical* to the partition function of the topological nonlinear σ model. Again, we have the important property

$$\frac{\delta Z_{\text{top}}^{d=2}}{\delta g_{\mu\nu}} = 0, \quad (3.7)$$

which is the analogue of Eq. (2.38). Moreover, the field content and all vacuum expectation values (VEV's) are the same in both theories.

Before coming to the evaluation of such VEV's let us again note that, instead of starting from $\mathcal{S}_{\text{top}}^{d=2}$, one could follow the Labastida-Pernici [23,26] approach and consider a classical Gaussian action

$$\mathcal{S}_G = \int (\mathbf{G} - \mathbf{F}[\mathbf{n}])^2 d^2x, \quad (3.8)$$

where $\mathbf{G}$ is an auxiliary field which transforms under variation (2.14) so as to render $\mathcal{S}_G$ invariant, i.e.,

$$\delta G = \frac{\delta \mathbf{F}}{\delta \mathbf{n}} \delta \mathbf{n}. \quad (3.9)$$

Again one has to fix this symmetry and the second-generation symmetry associated with rotations (2.27). If we chose the conditions

$$\mathbf{G} = 0, \quad \mathbf{c} \cdot \mathbf{n} = 0, \quad (3.10)$$

and integrate out the auxiliary field G , we again find a quantum action which this time coincides with that given

by Eqs. (2.36) and (2.37). Note that in this gauge, the equations of motion arising from action (3.8) are the instanton equations.

Since the TNLSM is a topological field theory, it is of interest to compute its topological invariants. In order to find operators $\mathcal{O}$ leading to topological invariants let us proceed as in Ref. [1] and look for those $\mathcal{O}$ satisfying the following conditions: (1) $\mathcal{O}$ is metric independent; (2) $\{Q, \mathcal{O}\} = 0$. If both properties are satisfied, then

$$\frac{\delta \langle \mathcal{O} \rangle}{\delta g_{\mu\nu}} = 0, \quad (3.11)$$

as can be easily seen if one proceeds as for deriving Eq. (2.40). Inspection of Eqs. (2.31)–(2.33) shows that, as usual, the ghost of ghost ϕ is a natural candidate for getting such invariants. Indeed, let us denote it as W_0 with the subscript zero indicating its zero-form character:

$$W_0 \equiv \phi(x). \quad (3.12)$$

Then, it is easy to show that

$$dW_0 = \{Q, W_1\}, \quad (3.13)$$

with W_1 a one-form given by

$$W_1 = \mathbf{n} \cdot d\mathbf{c}. \quad (3.14)$$

Now, one can see from Eq. (3.13) that $\langle W_0 \rangle$ is independent of the point x at which it is calculated. Indeed, integrating Eq. (3.13) one has

$$W_0(x_1) - W_0(x_0) = \int_{x_0}^{x_1} \{Q, W_1\} \quad (3.15)$$

and hence

$$\langle W_0(x_0) \rangle = \langle W_0(x_1) \rangle. \quad (3.16)$$

Now, we can repeat the procedure by noting that

$$dW_1 = \{Q, W_2\}, \quad (3.17)$$

with the two-form W_2 given by

$$W_2 = \frac{1}{2} \mathbf{n} \cdot (d\mathbf{n} \times d\mathbf{n}). \quad (3.18)$$

One recognizes in Eq. (3.18) the topological density whose integral over M gives the instanton number. (Note that $dW_2 = 0$.)

Apart from W_0 , other topological invariants can be constructed from the W_i 's by considering homology cycles γ_k on M and constructing integrals of the form

$$I[\gamma_k] = \int_{\gamma_k} W_k. \quad (3.19)$$

These integrals satisfy $\{Q, I\} = 0$ and depend only on the homology class of γ_k . For getting topological invariants one should consider products of the form

$$O_{\mathcal{S}} = \prod_{i=1}^s I[\gamma_k]. \quad (3.20)$$

Depending on the zero-mode structure associated with $\mathcal{S}_q^2$ these operators will give VEV's which correspond to topological invariants. In Sec. V we shall give details on how these invariants are calculated.

IV. CANONICAL QUANTIZATION AND THE EMERGENCE OF SPIN

In the preceding two sections we showed that the stochastic quantization of the theory of spin is equivalent to the path-integral quantization of the topological nonlinear σ model (TNLSM). In particular, we showed that the partition function of the TNLSM is *independent* of the coupling constant α of Eq. (3.1). This result is most surprising since the action Eq. (2.1) for the path integral for a degree of freedom with spin S depends explicitly on the spin. Somehow, the procedure of stochastic quantization seems to have erased all memory of the actual value of S . We will show in this section that the TNLSM contains *all* the possible realizations of spin, i.e., representations. In a way, the TNLSM is a *universal* theory of spin.

The resolution of this paradox will also take care of another apparent technical difficulty. In its standard form [16], the procedure of stochastic quantization is defined in terms of a Markov process generated by a Langevin equation. If the classical action (or Hamiltonian) is real, the process converges to an invariant equilibrium distribution which is the lowest eigenstate of the associated Liouville or Fokker-Planck operator. If we denote by $\mathcal{S}$ the classical action, the invariant distribution takes the standard Gibbs form $\exp(-\mathcal{S})$. Thus, it may appear that stochastic quantization cannot be used in theories with complex distributions. In the case of the path-integral quantization of spin the Feynman weight in the path integral (i.e., the distribution) is *always* complex, both in Minkowski and in Euclidean space. Thus, the use of stochastic quantization in this theory may be problematic. There is some degree of confusion on the role of the factor of i in the literature. It is usually argued that in these theories the presence of the factor of i in front of the action is only important if large gauge transformations have to be taken into account. We will see below that the canonical quantization of the TNLSM yields wave functions, which are the Feynman weights for the path-integral theory of spin, which include the *necessary* factor of i in the exponent.

Let us now discuss the canonical quantization of the TNLSM. Since we are dealing with a theory with constraints we will follow the Dirac procedure which imposes the constraints on the space of allowed states. We begin with the action for the TNLSM, Eq. (3.1):

$$\mathcal{S}_{\text{top}}^{d=2} = \alpha \int d^2x \epsilon_{\mu\nu} \mathbf{n} \cdot (\partial_\mu \mathbf{n} \times \partial_\nu \mathbf{n}) . \quad (4.1)$$

Following the procedure used by Horowitz [13] in his discussion of topological theories, we consider the equivalent action obtained by a Hubbard-Stratonovich transformation of Eq. (4.1). Let $\mathbf{B}_\mu$ be a vector field taking values on the tangent space of S^2 , i.e.,

$$\mathbf{n} \cdot \mathbf{B}_\mu = 0 . \quad (4.2)$$

Let $\mathcal{S}'$ be the action

$$\mathcal{S}' = 2\alpha \int_M d^2x \mathbf{n} \cdot (\partial_\mu \mathbf{n} \times \mathbf{B}_\nu - \frac{1}{2} \mathbf{B}_\mu \times \mathbf{B}_\nu) \epsilon_{\mu\nu} . \quad (4.3)$$

The equivalence of the path integrals defined by the actions of Eqs. (4.1) and (4.3) can be checked by an explicit

integration over the field $\mathbf{B}_\mu$. From the action $\mathcal{S}'$ we can derive the equations of motion

$$\mathbf{n} \times \epsilon_{\mu\nu} \partial_\mu \mathbf{B}_\nu = 0 , \quad (4.4a)$$

$$\partial_\mu \mathbf{n} = \mathbf{B}_\mu . \quad (4.4b)$$

The canonical procedure becomes more transparent if we write the action $\mathcal{S}'$ in terms of the components of the fields

$$\mathcal{S}' = 2\alpha \int_M d^2x [\partial_0 \mathbf{n} \cdot (\mathbf{B}_1 \times \mathbf{n}) - \mathbf{B}_0 \cdot (\mathbf{B}_1 \times \mathbf{n} + \mathbf{n} \times \partial_1 \mathbf{n})] . \quad (4.5)$$

Thus, $\mathbf{B}_0$ is a Lagrange multiplier field which enforces the constraint ("Gauss's law")

$$\mathbf{B}_1 \times \mathbf{n} + \mathbf{n} \times \partial_1 \mathbf{n} = 0 . \quad (4.6)$$

We can also read off the definition of the momentum $\boldsymbol{\pi}$ canonically conjugate to $\mathbf{n}$ from Eq. (4.5):

$$\boldsymbol{\pi} \equiv 2\alpha (\mathbf{B}_1 \times \mathbf{n}) , \quad (4.7)$$

where we have momentarily ignored the fact that, because of the constraint $\mathbf{n}^2 = 1$, there are only two independent degrees of freedom. If we now impose canonical commutation relations between the field $\mathbf{n}$ and the momentum $\boldsymbol{\pi}$, we find that we can define the operator

$$\mathbf{B}_1 \times \mathbf{n} = \frac{i}{2\alpha} \frac{\delta}{\delta \mathbf{n}} \quad (4.8)$$

which acts on the space of wave functions $\Psi[\mathbf{n}]$ which obey Gauss's law [Eq. (4.6)], namely

$$\left[\mathbf{n} \times \partial_1 \mathbf{n} - \frac{i}{2\alpha} \frac{\delta}{\delta \mathbf{n}} \right] \Psi[\mathbf{n}] = 0 . \quad (4.9)$$

Let us remark that the definition of the canonical momentum, Eq. (4.7), requires the weight of the path integral of the TNLSM to have the standard factor i in its exponent. Thus, in a sense, we have implicitly assumed that it is possible to do an analytic continuation of the coupling constant $\alpha \rightarrow i\alpha$.

In order to find the wave functions which solve the constraint we need to specify the quantization manifold. Locally (i.e., for each time slice) the manifold M can be viewed as the union of a set of circles times the real line, i.e., $S^1 \times \cdots \times S^1 \times \mathcal{R}$. Clearly, since canonical variables on disjoint circles commute with each other, the Hilbert space splits into as many disconnected factors as there are circles in the quantization manifold. We will restrict the rest of our discussion to the simpler case in which $M = S^1 \times \mathcal{R}$.

It is easy to check that the wave functions which satisfy Eq. (4.9) have the form

$$\Psi[\mathbf{n}] = \mathcal{N} \exp \left[-i2\alpha \int_0^1 d\lambda \int_0^T dx_1 \mathbf{n} \cdot (\partial_1 \mathbf{n} \times \partial_\lambda \mathbf{n}) \right] , \quad (4.10)$$

where the fields $\mathbf{n}$ are extensions from S^1 to the cap Σ of S^2 with boundary $\partial\Sigma = S^1$. $\mathcal{N}$ is a normalization constant. There are two possible extensions which corre-

spond to the two possible choices of caps. If we choose to include the north pole inside the cap, the natural extension is

$$\mathbf{n}(\lambda=0)=(1,0,0), \quad (4.11a)$$

$$\mathbf{n}(\lambda=1)=\mathbf{n}(x_1). \quad (4.11b)$$

The two choices of extensions lead to a change in the phase of the wave function by an amount of $\exp(-i2\alpha4\pi)$. Smoothness of the wave function on the circle S^1 demands that the coupling α should be restricted to take only the *quantized* values $\alpha=k/4$, where k is an integer. In the path-integral theory of a spinning particle of spin S , the spin is identified with the coefficient in front of the integral in the exponent of Eq. (4.10). Thus, we get $S=2\alpha=k/2$.

This is a very interesting result. First, it clearly shows that the wave function of the TNLSM or, what is the same, the invariant distribution, is exactly the same as

$$\langle k=2S|\{\mathbf{J}(t)\}\rangle = \int D\mathbf{n}(t) \exp \left[-iS \int_0^1 d\lambda \int_0^T dt \mathbf{n} \cdot (\partial_t \mathbf{n} \times \partial_\lambda \mathbf{n}) + \int dt \mathbf{J}(t) \cdot \mathbf{n}(t) \right]. \quad (4.13)$$

We can easily recognize this overlap as the generating function of the correlation functions of the theory of spin in path-integral form. We stress here that since these formulas are derived in canonical quantization, the “stochastic time” τ is fixed. Furthermore, since the states $|k\rangle$ are *invariant states*, in the sense that they are solutions of the constraint equations (i.e., they are “eigenstates” of the TNLSM) the overlaps, and hence the correlation functions, are *independent* of the choice of the quantization surface defined by τ . This result completes our argument on the equivalence of the two theories.

V. TOPOLOGICAL INVARIANTS

The topological nature of the $O(3)$ model that we have presented in the previous sections is summarized by Eq. (2.36) which tells us that *the complete action is a BRST transformation*. As a result, the partition function is independent of the metric and of the coupling constant. These crucial features will allow us to use a semiclassical expansion for the *exact* computation of VEV's.

Let $\mathcal{O}(\mathbf{n}, \mathbf{c}, \phi)$ be an arbitrary functional of the fields $\mathbf{n}, \mathbf{c}$, and ϕ . Because of the independence of Z on the coupling constant [Eq. (2.40)] we can evaluate $\langle \mathcal{O} \rangle$ in the weak-coupling limit where the path integral is dominated by classical minima. Hence $\langle \mathcal{O} \rangle$ is determined by instantons, solutions of Eq. (3.2). Now, it is well known [27] that when expanding around instanton solutions, it is necessary to take into account the fact that there exists in general an infinite number of solutions depending on certain parameters that we shall represent by a vector α . This vector spans a manifold $\mathcal{V}$ called the *moduli space*. When one performs a saddle-point expansion, these parameters manifest through the appearance of zero modes that have to be carefully handled. Following Ref. [27] we then write

the Feynman weight for the path integral for a spinning particle, with the correct factor of i in the exponent included. Second, despite the independence of the partition function and of all the topological invariants on the coupling constant α , the wave function not only depends on α but it requires its *quantization*. Third, all integer and half-integer values of the spin are admissible. Thus, the TNLSM is a universal theory of spin since it includes all the possible representations of $SU(2)$.

Finally, the wave function of the TNLSM can be used to compute the *correlation functions* of the theory of spin. Let us define by $|\{\mathbf{J}(t)\}\rangle$ the *coherent state*

$$|\{\mathbf{J}(t)\}\rangle = \exp \left[i \int dt \mathbf{J}(t) \cdot \mathbf{n}(t) \right] |0\rangle, \quad (4.12)$$

where $|0\rangle$ is the state of the TNLSM with $k=0$. We are interested in evaluating the *overlap* between the coherent state $|\{\mathbf{J}(t)\}\rangle$ and the state $|k=2S\rangle$, whose wave function is given by Eq. (4.10). The result is

$$\mathbf{n}(\mathbf{x}) \rightarrow \mathbf{n}_0(\mathbf{x}, \alpha) + \frac{1}{\alpha} \mathbf{n}_0(\mathbf{x}, \alpha) \times \boldsymbol{\epsilon}(\mathbf{x}), \quad (5.1)$$

$$\mathbf{c}(\mathbf{x}) \rightarrow \mathbf{c}_0(\mathbf{x}) + \frac{1}{\alpha} \mathbf{c}^\perp(\mathbf{x}), \quad (5.2)$$

$$\bar{\mathbf{c}}(\mathbf{x}) \rightarrow \frac{1}{\alpha} \bar{\mathbf{c}}(\mathbf{x}), \quad (5.3)$$

$$\phi(\mathbf{x}) \rightarrow -\frac{1}{2} [\mathbf{c}_0(\mathbf{x}) \times \mathbf{c}_0(\mathbf{x})] \cdot \mathbf{n}_0(\mathbf{x}) + \frac{1}{\alpha} \phi(\mathbf{x}), \quad (5.4)$$

$$\bar{\phi}(\mathbf{x}) \rightarrow \frac{1}{\alpha} \bar{\phi}, \quad (5.5)$$

where we have integrated out the Lagrange multipliers $\mathbf{b}$ and b_0 . Here $\mathbf{n}_0(\mathbf{x}, \alpha)$ is an instanton solution, $\mathbf{n}_0(\mathbf{x}, \alpha) \times \boldsymbol{\epsilon}(\mathbf{x})$ are the nonzero modes of $\delta^2(F^2)/\delta \mathbf{n}^2$, $\mathbf{c}_0(\mathbf{x})$ are the zero modes of operator $\mathcal{M}$, defined by Eq. (2.18), and $\mathbf{c}^\perp(\mathbf{x})$ are the remaining modes. The zero modes of $\mathcal{M}$ are given by

$$\mathbf{u}_i = \frac{\partial \mathbf{n}_0}{\partial \alpha_i} \times \mathbf{n}_0 \quad (5.6)$$

implying that the dimension of the kernel of $\mathcal{M}$ is the same as the dimension of the moduli space.

With all this, the integration measure, at lowest order in α , takes the form [27]

$$D\mathbf{n}(\mathbf{x}) = \prod_{i=1}^{\dim \mathcal{V}} A^i d\alpha^i D\boldsymbol{\epsilon}(\mathbf{x}), \quad (5.7)$$

$$D\mathbf{c}(\mathbf{x}) = \left[\prod_{i=1}^{\dim \mathcal{V}} \frac{1}{A^i} dc_0^i \right] D\mathbf{c}^\perp(\mathbf{x}). \quad (5.8)$$

The measures $D\bar{\mathbf{c}}(\mathbf{x}), D\phi(\mathbf{x}), D\bar{\phi}(\mathbf{x})$ are left unchanged. As usual, we have written

$$\mathbf{c}_0(\mathbf{x}) = \sum_{i=1}^{\dim \mathcal{V}} A^i \mathbf{u}_i(\mathbf{x}) c_i^0, \quad (5.9)$$

where c_i^0 are a set of Grassmann variables and A^i a normalization factor.

Now the (unnormalized) VEV $\langle \mathcal{O} \rangle$ can be expressed in the form

$$\langle \mathcal{O} \rangle = \int \left[\prod_i d\alpha^i d c_i^0 \right] D\epsilon D\mathbf{c}^\perp D\bar{\epsilon} D\phi D\bar{\phi} \times \exp(-\mathcal{S}) \mathcal{O}(\mathbf{n}_0, \mathbf{c}_0, -\frac{1}{2}\mathbf{n}_0 \cdot (\mathbf{c}_0 \times \mathbf{c}_0)) , \quad (5.10)$$

where

$$\mathcal{S} = \int \epsilon \mathcal{M}^\dagger \mathcal{M} \epsilon + \int \bar{\epsilon} \mathcal{M} \mathbf{c}^\perp + \int \bar{\phi} \phi + \int \mathbf{n}_0 \cdot (\mathbf{c}_0 \times \mathbf{c}^\perp) \bar{\phi} + O(1/\alpha) . \quad (5.11)$$

After performing the (trivial) integrations over ϕ and $\bar{\phi}$, we finally get

$$\langle \mathcal{O} \rangle = (\det' \mathcal{M}^\dagger \mathcal{M})^{-1/2} \det' \mathcal{M} \times \int \left[\prod_i d\alpha^i d c_i^0 \right] \mathcal{O}(\mathbf{n}_0, \mathbf{c}_0, -\frac{1}{2}\mathbf{n}_0 \cdot (\mathbf{c}_0 \times \mathbf{c}_0)) , \quad (5.12)$$

where $\det'$ means that zero modes are omitted when computing the determinant. Equation (5.12) shows that the only nontrivial VEV's are those corresponding to operators which can absorb the fermionic zero modes. Note that, as stressed in Ref. [1], we ended with an integral in the moduli space with a canonical measure

$$d\mu = \prod_i d\alpha^i \prod_i d c_i^0 \quad (5.13)$$

since $d\alpha^i$ and $d c_i^0$ transform oppositely under reparametrizations of $\mathcal{V}$.

Before an explicit computation of a topological invariant let us comment briefly about the instanton moduli space. For the $O(3)$ model the moduli space depends on the choice of the base space M and the degree of the field $\mathbf{n}(\mathbf{x})$ (i.e., its “topological charge”). For M a compact Riemann surface of genus g and $\mathbf{n}(\mathbf{x}):M \rightarrow S^2$ of degree $g \geq 2g$ the dimension of $\mathcal{V}$ is given by [28]

$$\dim \mathcal{V} = 4q - 2g + 2 . \quad (5.14)$$

As we explained in Sec. II our base space M is the compactification of $S^1 \times [-T, T]$ by choosing appropriate boundary conditions. We will consider as an example, M having the topology of the sphere. Then the field $\mathbf{n}(\mathbf{x})$ is a map,

$$\mathbf{n}(\mathbf{x}):S^2 \rightarrow S^2 , \quad (5.15)$$

and, in this case,

$$\dim \mathcal{V} = 4q + 2 . \quad (5.16)$$

As mentioned above, topological invariants are constructed from products of the operators W_i , $i=0,1,2$ defined in Eqs. (3.12), (3.14), and (3.18). One obtains non-vanishing values for those products that saturate the fermionic zero modes. Because of the triviality of the first homology group of $S^2[H_1(S^2, \mathbb{Z})=0]$, the only in principle nontrivial topological invariant is $\langle \phi^{2q+1}(\mathbf{x}) \rangle$.

To compute this VEV we use Eq. (5.12) with

$$\mathbf{n}_0(\mathbf{x}, \alpha), \quad \alpha = (\alpha^1, \dots, \alpha^{4q+2}) \in \mathcal{V} \quad (5.17)$$

a solution of the instanton equation [Eq. (3.2)] and the $4q+2$ zero modes of $\mathcal{M}$ given by

$$\mathbf{u}_i(\mathbf{x}, \alpha) = \frac{\partial \mathbf{n}_0(\mathbf{x}, \alpha)}{\partial \alpha^i} \times \mathbf{n}_0(\mathbf{x}, \alpha), \quad i = 1, \dots, 4q+2 . \quad (5.18)$$

It is useful to define the $(4q+2) \times (4q+2)$ real antisymmetric matrix:

$$\begin{aligned} \Delta_{ij}(\mathbf{x}, \alpha) &\equiv \frac{1}{2} \mathbf{n}_0 \cdot (\mathbf{u}_i \times \mathbf{u}_j) \\ &= \frac{1}{2} \mathbf{n}_0 \cdot \left[\frac{\partial \mathbf{n}_0(\mathbf{x}, \alpha)}{\partial \alpha^i} \times \frac{\partial \mathbf{n}_0(\mathbf{x}, \alpha)}{\partial \alpha^j} \right] . \end{aligned} \quad (5.19)$$

In terms of this matrix, the VEV reads

$$\langle \phi^{2q+1}(\mathbf{x}) \rangle = - \int_{\mathcal{V}} \prod_{i=1}^{4q+2} d\alpha^i \prod_{i=1}^{4q+2} d c_i^0 \sum_{i_1, j_1, \dots}^{4q+2} \Delta_{i_1 j_1} \cdots \Delta_{i_{2q+1} j_{2q+1}} c_{i_1}^0 c_{j_1}^0 \cdots c_{i_{2q+1}}^0 c_{j_{2q+1}}^0 \quad (5.20)$$

Performing the Berezin integrations we finally arrive at an integration on the moduli space of the $(2q+1)$ th power of a closed form:

$$\langle \phi^{2q+1}(\mathbf{x}) \rangle = - \int_{\mathcal{V}} (\Delta)^{2q+1} , \quad (5.21)$$

where

$$\Delta = \frac{1}{2} \sum_{i,j} \Delta_{ij} d\alpha_i \wedge d\alpha_j . \quad (5.22)$$

Notice that the integrand of Eq. (5.21) is nothing but the Pfaffian of the matrix Δ_{ij} , i.e.,

$$Pf(\Delta) = \sum_{i_1, j_1, \dots, i_{2q+1}, j_{2q+1}}^{4q+2} \epsilon_{i_1 j_1 \dots i_{2q+1} j_{2q+1}} \Delta_{i_1 j_1} \cdots \Delta_{i_{2q+1} j_{2q+1}} \quad (5.23)$$

which is proportional to the square root of the determinant of the matrix Δ .

Now we will prove the following lemma.

Lemma. Let $\mathbf{n}(\alpha)$ an arbitrary function $\mathbf{n}: \mathcal{V} \rightarrow \mathcal{R}^3$ and Δ_{ij} an antisymmetric matrix defined as in Eq. (5.18). Then for $q \leq 1$ the Pfaffian of Δ vanishes.

Proof. We have, for $q \geq 1$,

$$Pf(\Delta) = \sum_{a,b,c,d,\dots,i_{2q+1},j_{2q+1}}^{4q+2} \epsilon_{abcd} \dots \epsilon_{i_{2q+1}j_{2q+1}} \Delta_{ab} \Delta_{cd} \dots \Delta_{i_{2q+1}j_{2q+1}} \quad (5.24)$$

and we can write the tensor product $\Delta_{ab} \Delta_{cd}$ as

$$\begin{aligned} \Delta_{ab} \Delta_{cd} &= \frac{1}{4} \det \begin{pmatrix} n^1 & n^2 & n^3 \\ u_a^1 & u_a^2 & u_a^3 \\ u_b^1 & u_b^2 & u_b^3 \end{pmatrix} \det \begin{pmatrix} n^1 & u_c^1 & u_d^1 \\ n^2 & u_c^2 & u_d^2 \\ n^3 & u_c^3 & u_d^3 \end{pmatrix} \\ &= \frac{1}{4} \det \begin{pmatrix} \mathbf{n}^2 & \mathbf{n} \cdot \mathbf{u}_c & \mathbf{n} \cdot \mathbf{u}_d \\ \mathbf{u}_a \cdot \mathbf{n} & \mathbf{u}_a \cdot \mathbf{u}_c & \mathbf{u}_a \cdot \mathbf{u}_d \\ \mathbf{u}_b \cdot \mathbf{n} & \mathbf{u}_b \cdot \mathbf{u}_c & \mathbf{u}_b \cdot \mathbf{u}_d \end{pmatrix}. \end{aligned} \quad (5.25)$$

After expanding this determinant we see that each term in $\Delta_{ab} \Delta_{cd}$ is symmetric at least in one pair of indices $\{a,c\}, \{b,c\}, \{a,d\}$, or $\{b,d\}$. Because of the presence of the totally antisymmetric tensor $\epsilon_{abcd} \dots$ the contribution of each term to the Pfaffian is zero and

$$Pf(\Delta) = 0 \quad \text{for } q \geq 1. \quad (5.26)$$

The preceding lemma allows us to conclude that, in the case of the sphere, only the sector $q=0$ has a nonvanishing topological invariant. In this case, the solution of the instanton equation [Eq. (3.2)] is $\mathbf{n}_0$ a constant vector:

$$\mathbf{n}_0(\mathbf{x}; \alpha_1, \alpha_2) = \left[\frac{2\alpha_1}{1+\alpha_1^2+\alpha_2^2}, \frac{2\alpha_2}{1+\alpha_1^2+\alpha_2^2}, \frac{1-\alpha_1^2-\alpha_2^2}{1+\alpha_1^2+\alpha_2^2} \right]. \quad (5.27)$$

The computation of $\langle \phi(\mathbf{x}) \rangle$ leads to

$$\begin{aligned} \langle \phi(\mathbf{x}) \rangle &= - \int_{\mathcal{V}} \mathbf{n}_0 \cdot \left[\frac{\partial \mathbf{n}_0}{\partial \alpha_1} \times \frac{\partial \mathbf{n}_0}{\partial \alpha_2} \right] d\alpha_1 \wedge d\alpha_2 \\ &= -4 \int \frac{d\alpha_1 d\alpha_2}{(1+\alpha_1^2+\alpha_2^2)^2} = -4\pi. \end{aligned} \quad (5.28)$$

Then these results can be summarized in the following statement: *The only topological invariant of the $O(3)$ TNLSM when the base manifold is the sphere is $\langle \phi(\mathbf{x}) \rangle = -4\pi$.*

Let us finally remark that even if surprising, this result could have been anticipated by looking at the BRST structure of the invariants. In fact, it can be proved that for $q \geq 1$ the field $\phi^{2q+1}(\mathbf{x})$ is BRST exact:

$$\phi^{2q+1}(\mathbf{x}) = \left\{ Q, \frac{1}{2} \mathbf{c} \cdot (\mathbf{c} \times \mathbf{c}) \phi^{2q-1} + (\mathbf{c} \cdot \mathbf{n}) \phi^{2q} \right\}. \quad (5.29)$$

Thus, its VEV has to vanish.

VI. SUMMARY AND CONCLUSIONS

In this paper we have studied the connection between the path-integral theory of a spinning particle and the topological $O(3)$ nonlinear σ model. We have proven the

equivalence of the two theories in two different ways. Using the procedure of stochastic quantization on the theory of the spinning particle we showed that the generating functional of the stochastic theory is the same as the partition function of the TNLSM. We also showed that this partition function is independent of the coupling constant. The TNLSM is a topological QFT and, hence, its partition function is also independent of the metric. We also give a full classification of all the topological invariants described by the TNLSM.

The TNLSM was also quantized canonically. We showed that its wave functions are equal to the Feynman weights of the path-integral theory of spin. We further showed that the TNLSM contains all the possible representations of spin. We also showed that the spin-correlation functions are given by overlaps between the wave functions of the TNLSM and some properly chosen coherent states.

We have also calculated all the topological invariants of the TNLSM when the base manifold is the sphere. We found that all the invariants, except for one, vanish in all topological sectors. The only nonzero topological invariant is $\langle \phi(\mathbf{x}) \rangle = -4\pi$, where the nonvanishing contribution is due to the sector with zero topological charge.

Results of the sort that we derived here, were found originally in the framework of Chern-Simons and topological Yang-Mills theories. Its is interesting that a much simpler quantum-mechanical system, such as a spinning particle, and the TNLSM exhibit a structure very similar to that of their field-theoretic cousins. The path-integral theory of spin is an ideal laboratory for the study of the role of topology both in quantum mechanics and in quantum field theory.

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APPENDIX

In this appendix we summarize the formalism of coherent-state path integrals for spinning particles.

In the coherent-state quantization of the theory of a particle of spin S , first developed by Haldane [3] and later on rediscovered by many other authors [6,5,7,4], one be-

gins by defining a complete set of states $\{|S, M\rangle\}$ which span the $(2S+1)$ -dimensional representation of $SU(2)$ and are joint eigenstates of the generator J_3 and of the Casimir operator J^2 , i.e.,

$$J_3|S, M\rangle = M|S, M\rangle, \quad J^2|S, M\rangle = S(S+1)|S, M\rangle. \quad (A1)$$

Next, we define a set of coherent states $\{|\mathbf{n}\rangle\}$ by [29]

$$|\mathbf{n}\rangle \equiv e^{i\theta(\mathbf{n}_0 \times \mathbf{n} \cdot \mathbf{J})} |0\rangle, \quad (A2)$$

where the state $|0\rangle$ is the *highest-weight state* of the spin- S representation, i.e., $|0\rangle = |S, S\rangle$, $\mathbf{n}_0 = (1, 0, 0)$ and θ is the angle spanned by $\mathbf{n}_0$ and $\mathbf{n}$. The coherent states $|\mathbf{n}\rangle$ are defined up to a phase generated by rotations about $\mathbf{n}$ and form a Hermitian line bundle. The coherent states satisfy the identities

$$\langle \mathbf{n}_1 | \mathbf{n}_2 \rangle = e^{iS\Phi(\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_0)} \left(\frac{1 + \mathbf{n}_1 \cdot \mathbf{n}_2}{2} \right)^S, \quad (A3a)$$

$$\langle \mathbf{n} | \mathbf{J} | \mathbf{n} \rangle = S\mathbf{n}, \quad (A3b)$$

where $\Phi(\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_0)$ is the area of the spherical triangle with vertices at $\mathbf{n}_1, \mathbf{n}_2$, and $\mathbf{n}_0$. The quantization of the spin, $2S \in \mathbb{Z}$, makes harmless the 4π ambiguity in the definition of the area of the triangle. The coherent states $|\mathbf{n}\rangle$ also define a “resolution of the identity” $\hat{I}$,

$$\hat{I} = \int d\mu(\mathbf{n}) |\mathbf{n}\rangle \langle \mathbf{n}|, \quad (A4)$$

where $d\mu(\mathbf{n})$ is the invariant measure on the sphere S^2 , i.e.,

$$d\mu(\mathbf{n}) = \left[\frac{2S+1}{4\pi} \right] d^3\mathbf{n} \delta(\mathbf{n}^2 - 1). \quad (A5)$$

The set of coherent states $\{|\mathbf{n}\rangle\}$ can now be used to derive a path integral for the partition function for the spinning particle of the form [5]

$$Z_{\text{spin}} = \int \mathcal{D}\mathbf{n} \exp \left[iS \int_0^T dt \int_0^1 d\lambda \mathbf{n}(t, \lambda) \cdot [\partial_t \mathbf{n}(t, \lambda) \times \partial_\lambda \mathbf{n}(t, \lambda)] \right]. \quad (A6)$$

If the dynamics of the system is periodic, i.e., periodic boundary conditions in time, the histories of the field $\mathbf{n}$ sweep closed curves Γ on the sphere. The functions $\mathbf{n}(t, \lambda)$ are arbitrary extensions of the values taken by the field $\mathbf{n}$ on Γ to the cap Σ whose boundary is Γ . Once again, the ambiguity in the definition of the area is unob-

servable for the allowed values of S . This theory is topological in the sense that the action $S \times (\text{area of } \Sigma)$ is reparametrization invariant. This property is the analogue of the independence of the metric of the Chern-Simons theory.

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